

HUMAN GROWTH HORMONE

HGH / SOMATROPIN | 10IU 12IU 24IU 36IU | 4 VIAL SIZES

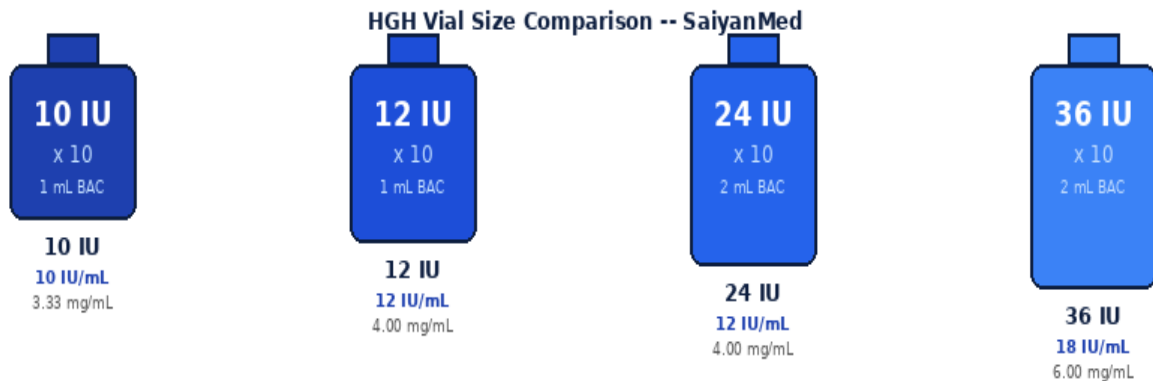
Recombinant Human Somatropin -- SubQ / IM Injection

Purity: 99.8% | Endotoxin-Free | Research Use Only

Human Growth Hormone (HGH / Somatropin) is a 191-amino acid peptide produced by recombinant DNA technology, identical in structure to endogenous pituitary GH. It requires reconstitution with Bacteriostatic Water before use and is administered by subcutaneous (SubQ) injection, preferably in the morning fasted. Available in four vial sizes: 10IU, 12IU, 24IU, and 36IU.

UNIT CONVERSION: 1 mg Somatropin = 3 IU | 1 IU = 0.333 mg | 1 IU = approximately 333 mcg

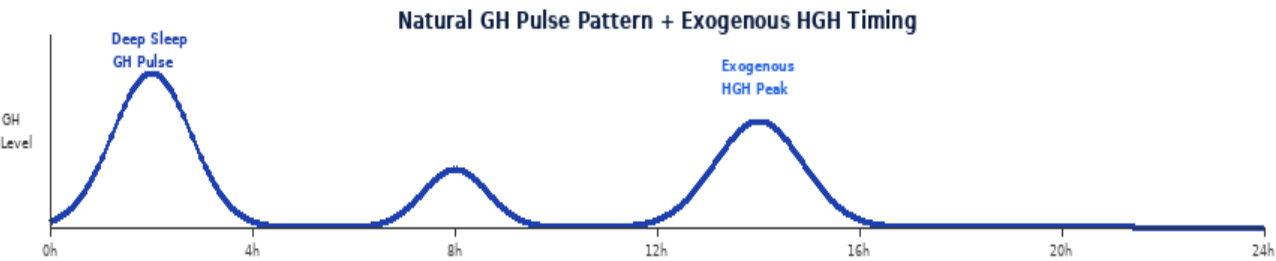
Vial Size Comparison



Vial	BAC Water	Concentration (IU/mL)	Concentration (mg/mL)	Doses per Vial (2IU ea)	Per Kit (x10)
10 IU x10	1 mL	10 IU/mL	3.33 mg/mL	5 doses	50 doses
12 IU x10	1 mL	12 IU/mL	4.00 mg/mL	6 doses	60 doses
24 IU x10	2 mL	12 IU/mL	4.00 mg/mL	12 doses	120 doses
36 IU x10	2 mL	18 IU/mL	6.00 mg/mL	18 doses	180 doses

MECHANISM OF ACTION & RESEARCH BENEFITS

Exogenous HGH (Somatropin) binds to GH receptors on target tissues, triggering a cascade of anabolic, lipolytic, and IGF-1-mediated effects. The pituitary normally releases GH in pulses — primarily during deep sleep and in response to exercise, fasting, and hypoglycaemia. Exogenous HGH supplementation augments these pulses, particularly during periods of natural trough levels.



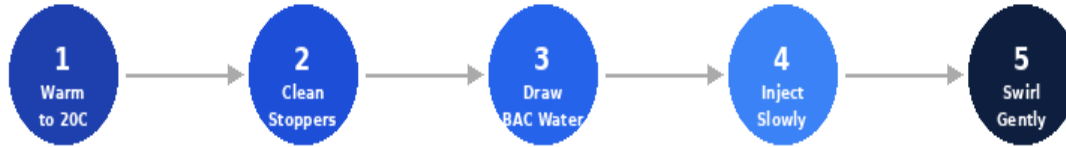
Pathway	Mechanism	Research Application
IGF-1 Stimulation	GH binds hepatic GH receptors; stimulates synthesis and secretion of IGF-1 (Insulin-like Growth Factor 1); IGF-1 mediates most anabolic effects of GH including protein synthesis and cell proliferation	Muscle hypertrophy, organ maintenance, tissue repair, bone density
Lipolysis	Direct GH receptor activation in adipocytes stimulates hormone-sensitive lipase (HSL); promotes fatty acid mobilisation from triglyceride stores; increases fat oxidation for energy	Body fat reduction, visceral fat clearance, improved body composition
Protein Synthesis	IGF-1 activates PI3K/Akt/mTOR pathway; enhances nitrogen retention; increases amino acid uptake into muscle cells; anti-catabolic (reduces muscle protein breakdown)	Lean mass preservation, muscle recovery, anti-aging research
Glucose Metabolism	GH has counter-insulin effects; reduces peripheral glucose uptake in muscle; increases hepatic glucose output; can cause insulin resistance at high doses	Metabolic research; requires monitoring blood glucose at higher doses
Collagen & Bone	Stimulates collagen synthesis in connective tissue; IGF-1 activates osteoblasts; increases bone mineral density; improves cartilage integrity	Joint health, bone density, skin elasticity, wound healing
Sleep & Cognition	GH and IGF-1 receptors in brain; promotes deep (slow-wave) sleep; neurogenesis support; neuroprotection	Sleep quality research, cognitive function, mood regulation

Key Research Benefits

Benefit	Evidence & Context
Body Composition	Research consistently shows reduction in fat mass and increase or preservation of lean mass at doses of 2-4 IU/day, with enhanced effects alongside resistance training.
Anti-Aging & Cellular Repair	GH and IGF-1 decline with age (somatopause). Supplementation restores anabolic and regenerative processes associated with younger physiology.
Recovery, Performance & Bone Density	Reduces recovery time after exercise and injury via enhanced collagen synthesis and protein turnover. Osteoblast stimulation via IGF-1 improves bone mineral density over 6-12 month protocols -- relevant in osteoporosis and age-related bone loss research.

RECONSTITUTION PROTOCOL

Always use Bacteriostatic Water (BAC Water) for reconstitution. HGH is a large, fragile protein — handle gently. Add BAC Water slowly and never inject directly onto the powder. Never shake. The 10IU and 12IU vials use 1 mL BAC Water. The 24IU and 36IU vials use 2 mL BAC Water.



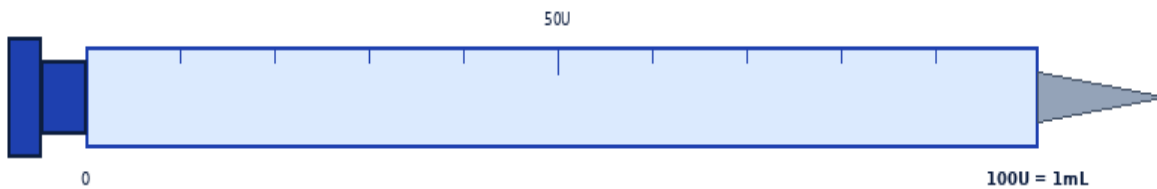
NEVER shake the vial -- roll gently. Solution must be clear before use.

Step	Action	Detail
1	Warm to room temperature	Remove vial and BAC Water from refrigerator. Allow both to reach ~20 deg C (room temperature) before reconstitution. HGH powder is delicate -- temperature shock during reconstitution can reduce potency.
2	Clean stoppers	Wipe rubber stoppers of BOTH vials with a fresh alcohol swab. Air-dry 10-15 seconds.
3	Draw BAC Water	Draw correct volume: 1 mL for 10IU or 12IU vials, 2 mL for 24IU or 36IU vials. Remove all air bubbles before proceeding.
4	Inject BAC Water into HGH vial	Direct the stream AGAINST THE GLASS WALL of the vial. NEVER inject directly onto the lyophilised powder -- this shears the GH protein and destroys potency.
5	Dissolve gently	Roll vial gently between palms. HGH may take 30-90 seconds to dissolve fully. Do NOT shake, vortex, or agitate vigorously. Solution must be clear and colourless before use.
6	Inspect	Discard if cloudy, particulate, or if solution has a colour. Slight foam at surface is normal and will settle.
7	Label and refrigerate	Write reconstitution date. Store at 2-8 deg C. Use within 28 days. Never freeze reconstituted HGH.

Reconstitution Quick Reference

Vial	BAC Water	IU/mL	mg/mL	1IU Volume	2IU Volume	4IU Volume
10 IU	1 mL	10	3.33	0.10 mL 10U	0.20 mL 20U	0.40 mL 40U
12 IU	1 mL	12	4.00	0.083 mL 8.3U	0.167 mL 16.7U	0.333 mL 33.3U
24 IU	2 mL	12	4.00	0.083 mL 8.3U	0.167 mL 16.7U	0.333 mL 33.3U
36 IU	2 mL	18	6.00	0.056 mL 5.6U	0.111 mL 11.1U	0.222 mL 22.2U

DOSE CALCULATION GUIDE



Insulin Syringe -- 1mL / 100U | Use 29-31G needle for SubQ

Formula: Volume (mL) = Desired Dose (IU) / Concentration (IU/mL) | **Units:** Volume (mL) x 100 = Units on insulin syringe

HGH Dose Range by Research Protocol



Vial: 10 IU x 10 -- Add 1 mL BAC Water → Concentration: 10 IU/mL (3.33 mg/mL)

Dose (IU)	1 IU	2 IU	3 IU	4 IU	6 IU	8 IU
Volume (mL)	0.100	0.200	0.300	0.400	0.600	0.800
Syringe Units	10.0U	20.0U	30.0U	40.0U	60.0U	80.0U

Vial: 12 IU x 10 -- Add 1 mL BAC Water → Concentration: 12 IU/mL (4.00 mg/mL)

Dose (IU)	1 IU	2 IU	3 IU	4 IU	6 IU	8 IU
Volume (mL)	0.083	0.167	0.250	0.333	0.500	0.667
Syringe Units	8.3U	16.7U	25.0U	33.3U	50.0U	66.7U

Vial: 24 IU x 10 -- Add 2 mL BAC Water → Concentration: 12 IU/mL (4.00 mg/mL)

Dose (IU)	1 IU	2 IU	3 IU	4 IU	6 IU	8 IU
Volume (mL)	0.083	0.167	0.250	0.333	0.500	0.667
Syringe Units	8.3U	16.7U	25.0U	33.3U	50.0U	66.7U

Vial: 36 IU x 10 -- Add 2 mL BAC Water → Concentration: 18 IU/mL (6.00 mg/mL)

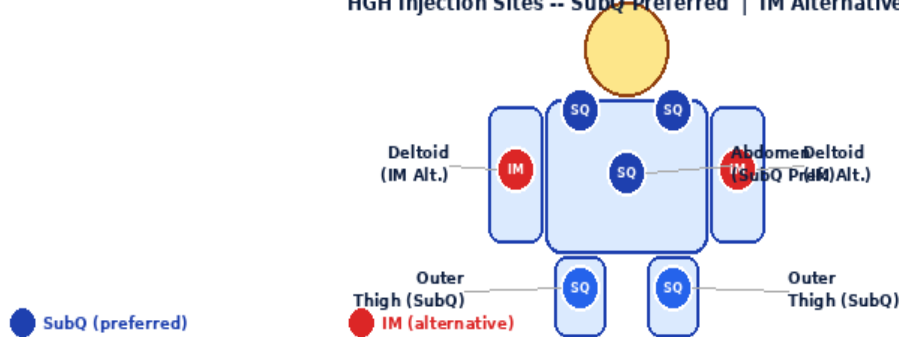
Dose (IU)	1 IU	2 IU	3 IU	4 IU	6 IU	8 IU
Volume (mL)	0.056	0.111	0.167	0.222	0.333	0.444
Syringe Units	5.6U	11.1U	16.7U	22.2U	33.3U	44.4U

INJECTION TECHNIQUE, SITES & SCHEDULE

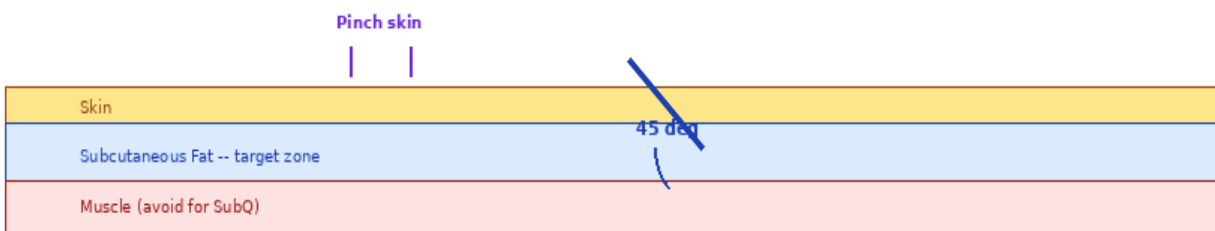
SubQ injection into abdominal fat is the preferred route for HGH — mimics natural pulsatile release most closely and minimises lipoatrophy risk. **IM injection** into the deltoid achieves faster peak plasma levels but is associated with more discomfort and potential lipoatrophy at injection sites. Inject **fasted in the morning** (30-60 min before eating) or at bedtime to align with the natural deep-sleep GH pulse.

Step	Action	Detail
1	Wash hands	Wash with soap and warm water 20 seconds. Dry completely.
2	Select syringe & needle	SubQ: 29-31G x 5/16 inch insulin syringe (1 mL). IM: 23-25G x 1 to 1.5 inch.
3	Draw dose carefully	Draw exact volume per dose table. Handle syringe gently. Expel air bubbles. HGH protein is fragile -- avoid agitation.
4	Clean injection site	Wipe site with alcohol swab. Allow to dry 10-15 seconds.
5	SubQ: pinch and inject	Pinch 2-3 cm of skin. Insert at 45 degrees. Release pinch. Inject slowly over 5-10 seconds.
6	Withdraw and press	Remove at same angle. Press with dry swab 5-10 sec. Do NOT rub.
7	Rotate site & dispose	Record injection site. Rotate systematically. Needle into sharps container.

HGH Injection Sites -- SubQ Preferred | IM Alternative



45-Degree SubQ Injection Angle



8-Point Abdomen Rotation Map



Research Dosing Protocols

Protocol	Daily Dose	Frequency	Duration	Key Notes
Anti-Aging / Wellness	1-2 IU	Daily	Ongoing (with breaks)	Lowest effective dose; minimal side effect risk; preserve sensitivity
Fat Loss & Recomp	2-4 IU	Daily	3-6 months	Morning fasted; may split AM + pre-bed; monitor glucose
Muscle Building (with anabolics)	4-6 IU	Daily	3-6 months	Add insulin sensitiser if using 4+ IU; advanced protocol
Performance Research	6-10 IU	Daily	12-24 weeks	High dose -- significant side effect risk; requires monitoring
Rest Period	0 IU	Off cycle	4-8 weeks minimum	Allows pituitary sensitivity to reset. Essential for long-term protocols.

Side Effects

Side Effect	Dose Relation	Management
Water retention / oedema	Dose-related	Common at 2+ IU. Reduce dose. Increase water intake. Resolves on reduction.
Carpal tunnel syndrome	Dose-related	Fluid accumulation in wrist. Reduce dose. Usually resolves within weeks.
Joint pain / arthralgia	Dose-related	Particularly wrists, knees. Reduce dose. Anti-inflammatory if needed.
Insulin resistance / high glucose	Dose-related	Monitor fasting blood glucose at 2+ IU. May require berberine or metformin.
Injection site lipoatrophy	Site-related	Fat loss at injection site with repetitive use. ROTATE sites rigorously.
Headache / fatigue	Common (early)	Usually transient in first 1-2 weeks. Reduce dose temporarily.
Acromegaly (at very high doses)	High dose only	Bone/organ growth. Only with multi-year, high-dose (10+ IU) abuse. Stay within research dose ranges.
IGF-1 elevation (tumour risk)	Dose-related	Contraindicated in any history of cancer. Monitor IGF-1 levels periodically.

STOP & SEEK MEDICAL ADVICE: Any history of cancer or active malignancy -- HGH is absolutely contraindicated

STOP & SEEK MEDICAL ADVICE: Severe or worsening carpal tunnel symptoms (numbness, weakness in hands)

STOP & SEEK MEDICAL ADVICE: Diabetic ketoacidosis or severely elevated blood glucose

STOP & SEEK MEDICAL ADVICE: Sudden severe headache, vision changes, or pituitary apoplexy symptoms

STOP & SEEK MEDICAL ADVICE: Signs of injection site infection: spreading redness, warmth, pus, fever

Absolute Contraindications: Active cancer or any malignancy history, active proliferative or severe non-proliferative diabetic retinopathy, acute critical illness (trauma, open heart surgery, respiratory failure), Prader-Willi syndrome with obesity/airway obstruction, pregnancy or lactation. Not for use in individuals under 18.

Storage Guidelines

State	Temperature	Duration	Notes
Unreconstituted powder	2-8 deg C	Per expiry	Keep upright. Away from light. Do not freeze. Some sources allow short room-temp storage -- always prefer fridge.
Reconstituted solution	2-8 deg C	28 days	Label with reconstitution date. Inspect before each use. Slight cloudiness on draw = warm to room temp and swirl gently.
BAC Water (opened)	2-8 deg C	28 days	Label opening date.
Transport	Below 25 deg C	Max 72 hr	Insulated bag with ice packs. Never freeze. Avoid direct ice contact.

Golden Rules -- HGH

1. NEVER shake the HGH vial -- HGH is a fragile protein; agitation destroys potency.
2. Direct BAC Water against the glass wall, never onto the powder.
3. Use 1 mL BAC Water for 10IU and 12IU vials; 2 mL for 24IU and 36IU vials.
4. Inject fasted in the morning (30-60 min before eating) for optimal fat-loss effect.
5. Rotate SubQ injection sites -- lipoatrophy occurs if same site is used repeatedly.
6. Monitor fasting blood glucose at 2+ IU/day -- HGH causes insulin resistance.
7. Start low (1-2 IU/day) and escalate gradually over 2-4 weeks.
8. Store reconstituted HGH at 2-8 deg C. Never freeze. Discard after 28 days.
9. Absolutely contraindicated in any history of cancer -- HGH stimulates IGF-1 and cell growth.
10. Take mandatory off-cycle breaks (4-8 weeks minimum) to preserve pituitary sensitivity.

DISCLAIMER: This document is produced by _____ for informational and research reference purposes only. Human Growth Hormone (Somatropin) is a controlled prescription substance in many jurisdictions and a research compound in this context. This guide does not constitute medical advice. HGH must not be used therapeutically without a valid prescription from a licensed physician. _____ assumes no liability for misuse of this product or information herein.

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